

Jérôme Botoko Ekila

ACADEMIC CV

Pleinlaan 9, 1050 Brussels, Belgium

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Education

Vrije Universiteit Brussel

PHD IN SCIENCES

- Promotors: Prof. dr. Paul Van Eecke & Prof. dr. Katrien Beuls
- Research on emergent communication and construction grammar

Brussels, Belgium

Nov 2021 - present

Vrije Universiteit Brussel

MASTER OF SCIENCE IN APPLIED SCIENCES & ENGINEERING: COMPUTER SCIENCE

- Honor: magna cum laude (84%)
- Profile: Artificial Intelligence
- Thesis: Neuro-symbolic Procedural Semantics for Visual Question Answering

Brussels, Belgium

2018 - 2020

Vrije Universiteit Brussel

BACHELOR OF SCIENCE: COMPUTER SCIENCE

- Honor: magna cum laude (78%)

Brussels, Belgium

2016 - 2019

Research Experience

Sorbonne Université - Institut des Systèmes Intelligents et de Robotique

VISITING RESEARCHER

- Advisors: Prof. dr. Nicolas Bredeche
- Research on emergent communication and swarm robotics

Paris, France

Nov. 2025 - Jan. 2026

Vrije Universiteit Brussel - Artificial Intelligence Laboratory

RESEARCH ENGINEER

- Developed two web applications demonstrating novel research from the EHAL research group
- Developed a neuro-symbolic AI system integrating linguistic and visual reasoning
- Assisted PhD researchers by analyzing and reproducing novel state-of-the-art methodologies

Brussels, Belgium

Nov. 2020 - Oct. 2021

Publications

JOURNALS

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2026). Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. *Transactions of the Association for Computational Linguistics (TACL)* 14 (pp. 2234-2257). doi: 10.1162/TACL.a.799

Jamie Wright, Verheyen, L., De Ponte, F., **Botoko Ekila, J.**, Beuls, K. & Van Eecke, P. (2026). Conventionality Produces More Robust, Learnable and Efficient Emergent Languages. *Journal of Language Evolution* 11 (p. lzag006). doi: 10.1093/jole/lzag006

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2025). Neuro-Symbolic Procedural Semantics for Explainable Visual Dialogue. *PLOS One* 20(5) (p. e0323098). doi: 10.1371/journal.pone.0323098

Van Eecke, P., Beuls, K., **Botoko Ekila, J.**, & Rădulescu, R. (2022). Language games meet multi-agent reinforcement learning: A case study for the naming game. *Journal of Language Evolution* 7(2) (pp. 213-223). doi: 10.1093/jole/lzad001

CONFERENCES

Toquebiau M., **Botoko Ekila, J.**, Beuls, K. & Van Eecke, P. (Accepted). Cultural Evolution Drives the Emergence of Conventions in Neural Emergent Communication. *EMNLP 2026*

Botoko Ekila, J., Nevens, J., Verheyen, L., Beuls, K. & Van Eecke, P. (2024). Decentralised Emergence of Robust and Adaptive Linguistic Conventions in Populations of Autonomous Agents Grounded in Continuous Worlds. *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024)* (pp. 2168-2170).

Botoko Ekila, J. (2024). Emergence of Linguistic Conventions in Multi-Agent Systems Through Situated Communicative Interactions. *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024)* (pp. 2725–2727).

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2023). Neuro-Symbolic Procedural Semantics for Reasoning-Intensive Visual Dialogue Tasks. *26th European Conference on Artificial Intelligence (ECAI 2023)* (pp. 2419-2426). doi: 10.3233/FAIA230544

WORKSHOPS

Botoko Ekila, J., Verheyen, L., Beuls, K. & Van Eecke, P. (2025). Constructions All the Way Up: From Sensory Experiences to Construction Grammars. *Proceedings of the Second International Workshop on Construction Grammars and NLP (CxGs+NLP 2025)* (pp. 84-95)

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2022). Hybrid Procedural Semantics for Visual Dialogue: An Interactive Web Demonstration. In *Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJCAI 2022): Workshop on semantic techniques for narrative-based understanding* (pp. 48-52).

IN PREPARATION

Cazenille L., Macabre L., **Botoko Ekila, J.**, Bredeche, N. (In prep.). Pogotim – a Simulator for Pogobot robots.

Botoko Ekila, J., Toquebiau M., Cazenille L., Bredeche, N. (In prep.). The interplay of the evolutionary landscape and emergent languages.

Peer-reviewed abstracts (Presentations)

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2026). Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. TACL paper accepted for oral presentation (presented by Jérôme Botoko Ekila) *Empirical Methods in Natural Language Processing (EMNLP 2026)*, Budapest, Hungary.

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2026). Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. Encore abstract accepted for oral presentation (presented by Jérôme Botoko Ekila) *Belgium-Netherlands Conference on Artificial Intelligence (BNAIC/BENELEARN 2026)*, Maastricht, Belgium.

Toquebiau M., **Botoko Ekila, J.**, Beuls, K. & Van Eecke, P. (2026). Cultural Evolution Drives the Emergence of Conventions in Neural Emergent Communication. Abstract accepted for oral presentation (presented by Maxime Toquebiau). *Computational Linguistics in the Netherlands 36 (CLIN 36)*, Brussels, Belgium.

Jamie Wright, Verheyen, L., De Ponte, F., **Botoko Ekila, J.**, Beuls, K. & Van Eecke, P. (2025). Conventionality Produces More Robust, Learnable and Efficient Emergent Languages. Abstract accepted as poster. *Computational Linguistics in the Netherlands 35*, Leuven, Belgium.

Botoko Ekila, J., Verheyen, L., Beuls, K. & Van Eecke, P. (2025). Constructions All the Way Up: From Sensory Experiences to Construction Grammars. Abstract accepted as poster. *Computational Linguistics in the Netherlands 35 (CLIN 35)*, Leuven, Belgium.

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2024). Decentralised Emergence of Robust and Adaptive Linguistic Conventions in Populations of Autonomous Agents Grounded in Continuous Worlds. Accepted for oral presentation (presented by Jérôme Botoko Ekila). *Adaptive and Learning Agents Workshop @ AAMAS (ALA 2024)*, Auckland, New-Zealand.

Van Eecke, P., Beuls, K., **Botoko Ekila, J.** & Rădulescu, R. (2023). Investigating emergent communication through language games and multi-agent reinforcement learning. Encore abstract accepted for oral presentation (presented by Jérôme Botoko Ekila). *Belgium-Netherlands Conference on Artificial Intelligence (BNAIC/BENELEARN 2023)*, Delft, Belgium.

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2023). Decentralised Emergence of Robust and Adaptive Linguistic Conventions in Populations of Autonomous Agents Grounded in Continuous Worlds. Abstract accepted as poster. *Computational Linguistics in the Netherlands 33 (CLIN 33)*, Antwerp, Belgium.

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2022). Hybrid Procedural Semantics. Abstract accepted for oral presentation (presented by Lara Verheyen). *Computational Linguistics in the Netherlands 32 (CLIN 32)*, Tilburg, Netherlands.

Van Eecke, P., Beuls, K., **Botoko Ekila, J.** & Rădulescu, R. (2022). Investigating emergent communication through language games and multi-agent reinforcement learning. Abstract accepted as poster. *Computational Linguistics in the Netherlands 32 (CLIN 32)*, Tilburg, Netherlands.

Invited Talks

Mar. 2025. Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. Invited talk at Nicolas Bredeche's lab, Institute for Intelligent Systems and Robotics (ISIR), Sorbonne Université, Paris, France.

Teaching Experience

2027	Course: Natural Language Processing Teaching Assistant, 6 ECTS	Prof. dr. Paul Van Eecke
2023-2027	Course: Introduction to Computational Linguistics Teaching Assistant, 3 ECTS	Prof. dr. Paul Van Eecke
2023-2027	Course: Advanced Topics in Natural Language Processing Teaching Assistant, 6 ECTS	Prof. dr. Paul Van Eecke
2022-2027	Course: Wetenschappelijk Rekenen Teaching Assistant, 6 ECTS	Prof. dr. Kurt Barbé, Prof. dr. Ann Nowé
2024	Course: Artificial Intelligence Paradigms Teaching Assistant, 6 ECTS	Prof. dr. Paul Van Eecke
2022	Course: Artificial Intelligence Teaching Assistant, 6 ECTS	Prof. dr. dr. Geraint Wiggins

Academic Service and Community Engagement

REVIEWING

2025-2026	Benelux Conference on Artificial Intelligence (BNAIC)
2024-2026	Adaptive and Learning Agents - Workshop at AAMAS
2025	International Workshop on Construction Grammars and NLP (CxGs+NLP)
2023	European Conference on Artificial Intelligence (ECAI)

SERVICE

2026	Organising committee - Evolutionary Linguistics Association (ELA) Symposium	Paris, France
2026	Organising committee - Computational Linguistics in the Netherlands (CLIN36)	Brussels, Belgium
2023	Volunteer - Belgium NLP Meetup 2023	Brussels, Belgium

Awards, Fellowships, & Grants

July 2025	Mobility Grant for long stay abroad (3 months) Research Foundation Flanders (FWO)
July 2025	Mobility Grant for participation in a workshop Research Foundation Flanders (FWO)
May 2024	Student Scholarship AAMAS International Foundation for Autonomous Agents and Multiagent Systems (IFAAMAS)

Mentoring

- June 2026 **Jitse De Beer** Bachelor Thesis - *Pre-linguistic Category Formation in Grounded Language Games*, Vrije Universiteit Brussel
- June 2026 **Farah Ayyad** Master Thesis - *Concept Representation and Lexical Convergence in Grounded Language Games*, Vrije Universiteit Brussel
- Sep. 2025 **Xantin Présér** Master Thesis - *Grounded concepts in multi-word utterances: A methodology for the emergence of compositional languages*, Vrije Universiteit Brussel & UNamur
- June 2025 **Maël-Questiaux** Master Thesis - *Language games in Virtual Reality*, Vrije Universiteit Brussel & UNamur
- Sep. 2024 **Khalil Ouald Chaib** Master thesis - *LLM Detection: PCFG-based authorship attribution for detecting LLM-generated texts*, Vrije Universiteit Brussel